



Experimental Zika Vaccine

Researchers will be testing a new experimental vaccine for the Zika virus on humans for the first time.
<http://dgit.in/zikavaccine>



Gravitational Waves Again

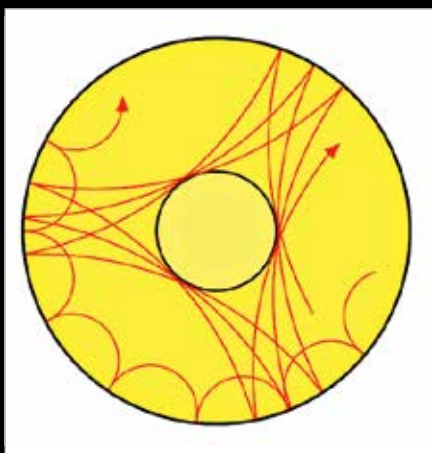
The scientists at LIGO have claimed to have detected gravitational waves a second time.
<http://dgit.in/gravitwave>

Space Age

convective layers. These convective layers are home to large scale turbulence that generate acoustic waves in a large range of frequencies. These waves drive random stellar oscillations and occur in stars that have masses and evolutionary patterns similar to the Sun. Since these stars are intrinsically stable these random oscillations have small amplitude and generally short lived.

Ok. So now we know what stellar oscillations are and how stars oscillate, but how does asteroseismology use this information?

Because sound cannot propagate in space, sound waves remain trapped inside



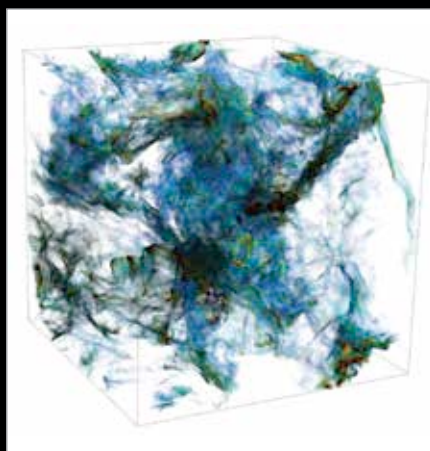
Owing to differential sensitivity of modes we can observe different regions in a star

a star. But as we discussed these sound waves cause a star to pulsate, oscillate in a combination of modes. These oscillations provide us with an indirect way to listen to the sounds of star. As the surface of a star recedes and expands the gas inside is compressed and then rarefied in the process getting hotter and then colder. These changes are then reflected in the light emitted by the stars. Thus, by measuring the light output of these pulsating stars we can gauge the frequencies of oscillations and the sound waves inside the star.

But, now that we have the frequency of oscillations and the frequency of sound how do we map the internal structure of a star with that information? One of the primary goals of asteroseismology is to

measure the speed of sound waves in a star. Sound waves propagate longitudinally. In a gas this results in information transfer by compression and rarefaction events, via collisions at the molecular level. This is important as it highlights the dependence of the speed of sound on how frequently molecules collide, which in turn is determined by temperature. The higher the temperature of a gas, the more mobile and energetic are the molecules and this increases the speed of sound. At a given temperature, a lighter gas moves more quickly than a more dense one. Speed of sound is thus critically related to the density and temperature of a gaseous medium. Using observation techniques like spectroscopy and photometry scientists can measure sound wave data within a star. Then, by making use of standard models, the wave data can be interpreted to give details of speed of sound waves, which in turn can be used to gauge density and pressure of the gas medium. Arguably the most important physical relation in thermodynamics is the equation of state, which relates the temperature of a state to its density and pressure. Armed with data of pressure and density of the gaseous medium that fills up the star, along with some assumptions made of the chemical composition of the gas, a plot of temperature within the star can be derived.

The many modes of oscillations also have importance in decoding a star's internal structure. Waves dominating the interior of a pulsating star can be broadly



A high fidelity numerical simulation showing the formation of a star (Credit : NASA)

classified into two modes - the p mode that stems from pressure as a restoring force and g modes that originate from gravity behaving as a restoring force for stellar oscillations. Different modes penetrate to different depths of the interior owing to difference in their nature, much like the s and p waves discussed in the context of earth based seismology. Because waves of a certain frequency and propagation direction might be sensitive to the physical conditions prevailing in



Much like an orchestra with different instruments a star oscillating with different modes can be 'heard'

different regions of the star's interior, by studying different and more pulsating modes we can cover a larger area of the star's inner structure.

Can we actually 'hear' stars?

Music is essentially a combination of a large number of instruments emitting sound waves of a certain frequency and amplitude. If we stand in a concert hall with a famous orchestra playing their famous score we can make out every individual instrument, while also appreciating the combined effect of music as a whole. Our brain is attuned to splitting a signal into its constituent waves and recognizing a unique sound, as it is capable of enjoying the power of music. When a star oscillates, its surface beats in a unique rhythm, driven by the breathing in-out of the sound trapped inside. While the natural frequency of stars is too low for humans to perceive the sound, when we capture the frequency, amplitude and phases associated with stellar oscillations, we can ramp up the necessary parameters to fall inside our audible range, thus listening in on the music of stars. Here's an audio byte: <http://dgit.in/Stlrr>